Assumptions used in Monte Carlo lattice pricing model	Year Ended December 31, 2016	2015	2014
Risk-free interest rate	**1.0%**	1.1%	0.7%
Expected dividend yield	**—%**	—%	—%
Expected volatility—ANSYS stock price	**21%**	23%	25%
Expected volatility—NASDAQ Composite Index	**16%**	14%	15%
Expected term	**2.8 years**	2.8 years	2.8 years
Correlation factor	**0.65**	0.60	0.70

The Company issued 35,000, 115,485 and 39,900 performance-based restricted stock awards during 2016, 2015 and 2014, respectively. Of the cumulative performance-based restricted stock awards issued, defined operating metrics were assigned to 63,462, 51,795 and 20,667 awards with grant-date fair values of $84.61, $86.38 and $81.52 during 2016, 2015 and 2014, respectively. The grant-date fair value of the awards is being recorded from the grant date through the conclusion of the measurement period associated with each operating metric based on management's estimates concerning the probability of vesting. As of December 31, 2016, 7,625 units of the total 2014 awards granted were earned and will be issued in 2017. Total compensation expense associated with the awards recorded for the years ended December 31, 2016, 2015 and 2014 was $0.4 million, $0.4 million and $0.1 million, respectively.

In addition, in 2016, 2015 and 2014, the Company granted restricted stock units of 488,622, 344,500 and 364,150, respectively, that will vest over a three- or four-year period with weighted-average grant-date fair values of $88.51, $86.34 and $82.13, respectively. During 2016 and 2015, 162,019 and 85,713 shares vested and were released, respectively. As of December 31, 2016, 2015 and 2014, 838,327, 571,462 and 344,750 units were outstanding, respectively. Total compensation expense is being recorded over the service period and was $19.1 million, $12.5 million and $5.8 million for the years ended December 31, 2016, 2015 and 2014, respectively.

In conjunction with a 2015 acquisition, ANSYS issued 68,451 shares of replacement restricted stock with a weighted-average grant-date fair value of $90.48. Of the $6.2 million grant-date fair value, $3.5 million, related to partially vested awards, was recorded as non-cash purchase price consideration. The remaining fair value will be recognized as stock compensation expense through the conclusion of the service period. During the years ended December 31, 2016 and 2015, the Company recorded $1.2 million and $0.6 million, respectively, of stock compensation expense related to these awards.

In conjunction with a 2011 acquisition, the Company granted performance-based restricted stock awards. Vesting was determined based on the achievements of certain revenue and operating income targets of the entity post-acquisition. Total compensation expense associated with the awards recorded for the year ended December 31, 2014 was $4.7 million.

The Company has granted deferred stock awards to non-affiliate Independent Directors, which are rights to receive shares of common stock upon termination of service as a Director. In 2015 and prior, the deferred stock awards were granted quarterly in arrears and vested immediately upon grant. Associated with these awards, the Company established a non-qualified 409(a) deferred compensation plan with assets held under a rabbi trust to provide Directors an opportunity to diversify their vested awards. During open trading windows and at their elective option, the Directors may convert their Company shares into a variety of non-Company-stock investment options in order to diversify their holdings. As of December 31, 2016, 5,000 shares have been diversified and 184,099 undiversified deferred stock awards have vested with the underlying shares remaining unissued until the service termination of the respective Director owners. In May 2016, the Company granted 38,400 deferred stock awards which will vest in full on the one-year anniversary of the grant. Total compensation expense associated with the awards recorded for the years ended December 31, 2016, 2015 and 2014 was $1.9 million, $4.0 million and $3.5 million, respectively.